

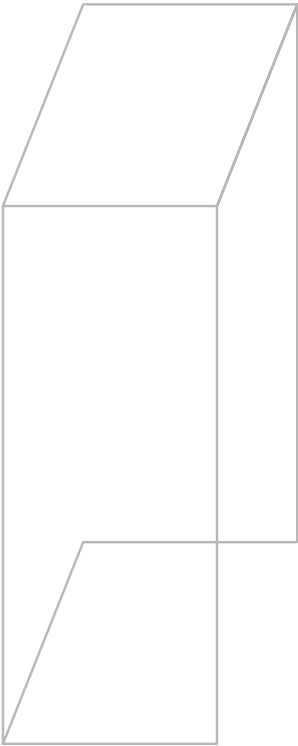
Cross-Cube Vega Hedging

A Swaption Market-Making Lab

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Independent research - produced as a portfolio artefact

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Abstract

A swaption market-maker sees vega by cube point, but the realised cube does not move one point at a time. It moves through correlated factors: ATM level, expiry slope, curvature,

and skew-wing dynamics.

This lab builds a simulated cube, calibrates SABR smiles, runs a client-flow market-maker,

and compares four hedging policies. The factor-neutral policy is not magic: it pays more hedge cost and relies on liquid proxy instruments. It does, however, make the residual

cross-cube P&L smaller and easier to explain.

Executive Summary

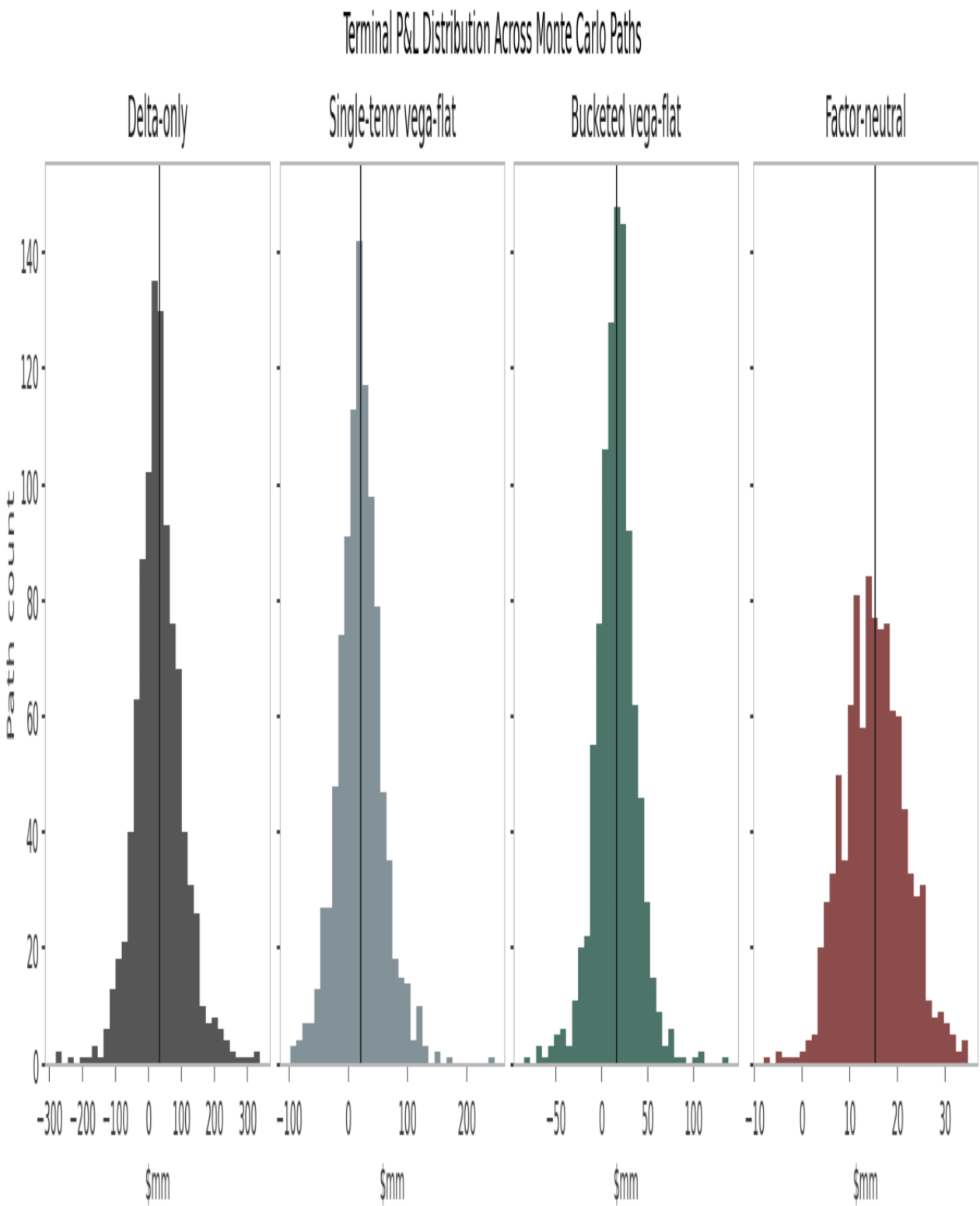
The headline result is mechanical and useful: hedging raw node vega can leave the book long

one cube factor and short another. Bucketed hedges reduce the leak. Factor-neutral hedges

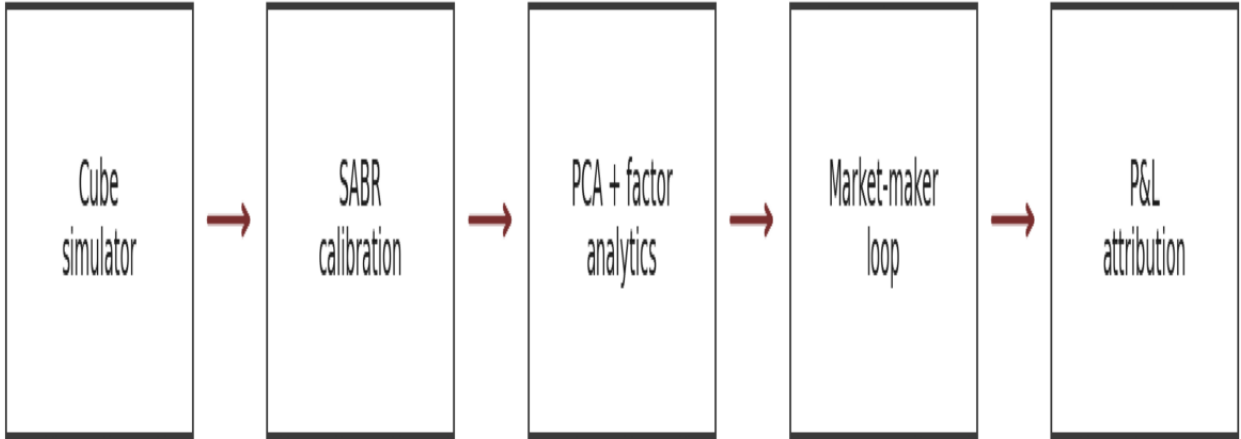
reduce it further, especially in the left tail.

The experiment uses no real market data. Parameter choices are set to published stylised

facts and kept explicit in YAML so the reader can change them and rebuild the note.



Architecture Diagram



One seed rebuilds every path, table, figure, HTML page, and PDF report.

Simulator to calibration to analytics to market-making loop to attribution.

Introduction

Node vega is operationally convenient. Traders quote, risk systems aggregate, and hedgers

choose liquid points. But the reporting basis is not the risk basis.

A client trade in a 2y into 10y payer does not expose the book only to the 2y x 10y mark. It

loads on the vol level, on the front-back slope, on the hump in intermediate expiries, and

on smile dynamics. This is where naive hedges leak P&L.

Architecture

The project is deliberately modular. The simulator creates the true cube. Calibration

recovers per-slice and constrained SABR parameters. Analytics recover factors. The market-

maker loop creates inventory, the hedger trades proxy instruments, and attribution

reconciles the realised P&L.

Model: True Cube

ATM log-vol is modelled as $\mu(T, \tau)$ plus three smooth loadings times OU factors. The

loadings are level, log-expiry slope, and a humped curvature mode.

SABR beta is fixed. Rho follows a bounded OU process in $[-0.95, 0.0]$. Log-nu follows OU

dynamics. Both receive shocks correlated with the level factor so skew steepens when

volatility rises.

Stylised Facts

The front of the normal-vol cube is set around 60-90bp and the back is anchored near 80bp.

The exact values are not calibrated to a live market; they are a plausible laboratory scale.

The report cites Andersen-Piterbarg, Hagan et al., Rebonato, and Bergomi for interest-rate

volatility modelling conventions, SABR smile dynamics, and volatility-surface factor

behaviour.

Calibration

Per-slice calibration follows market practice: beta fixed, ATM vol matched exactly, and rho

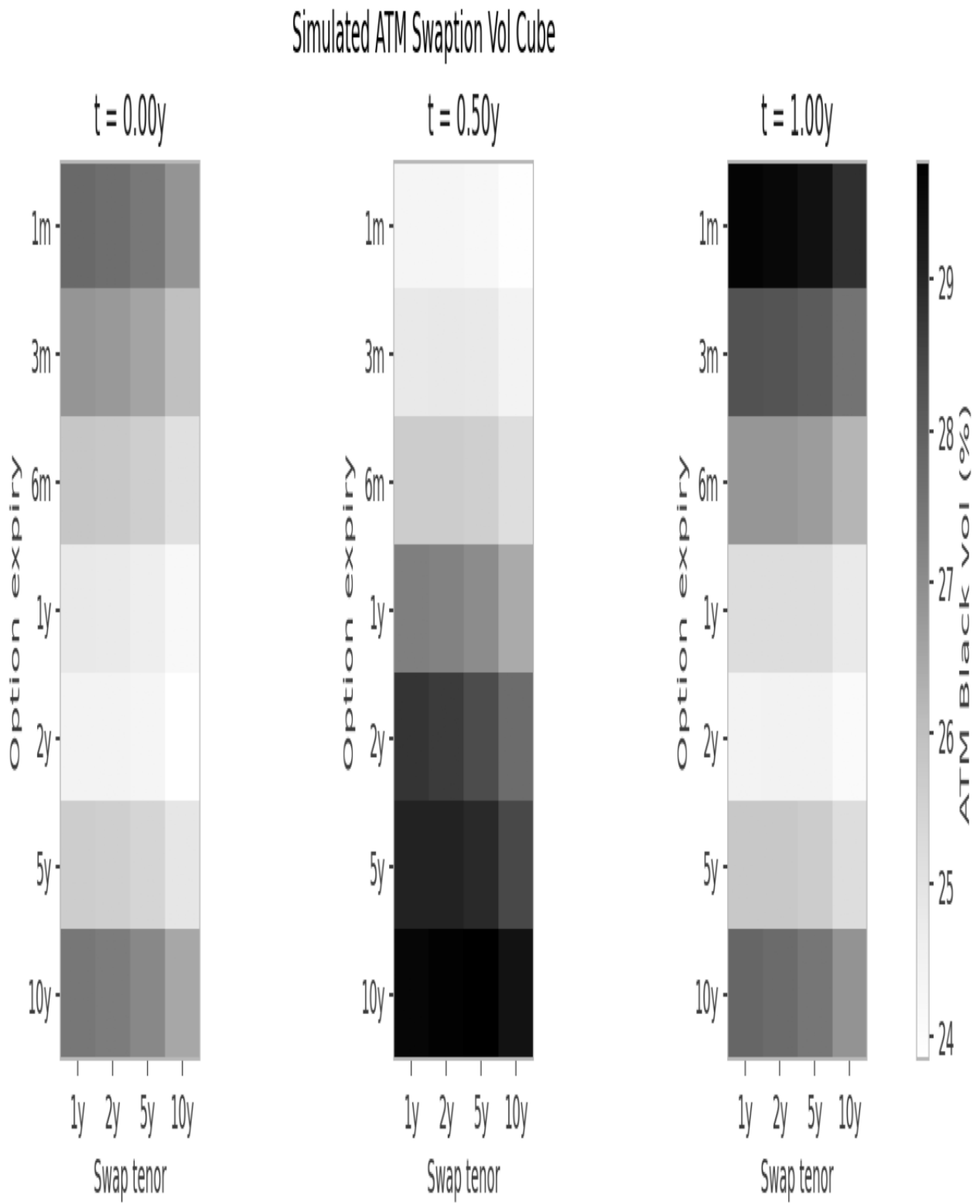
and nu fitted to the smile. The joint fit adds smoothness and static-arbitrage penalties.

When the joint fit sacrifices a small amount of local residual to reduce cube-level

roughness, that is not hidden. The residual heatmaps and violation count table expose the

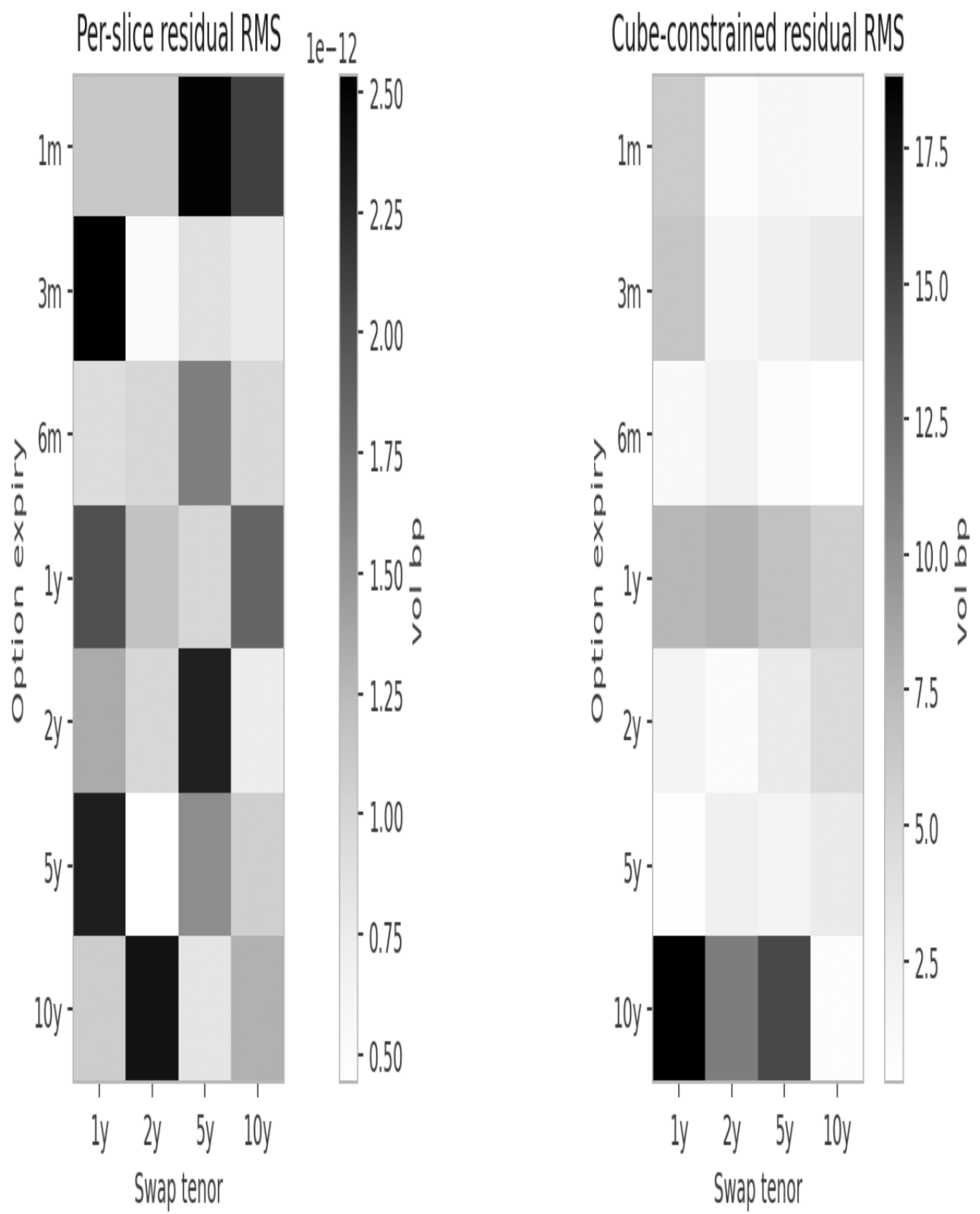
trade-off.

ATM Cube Snapshots



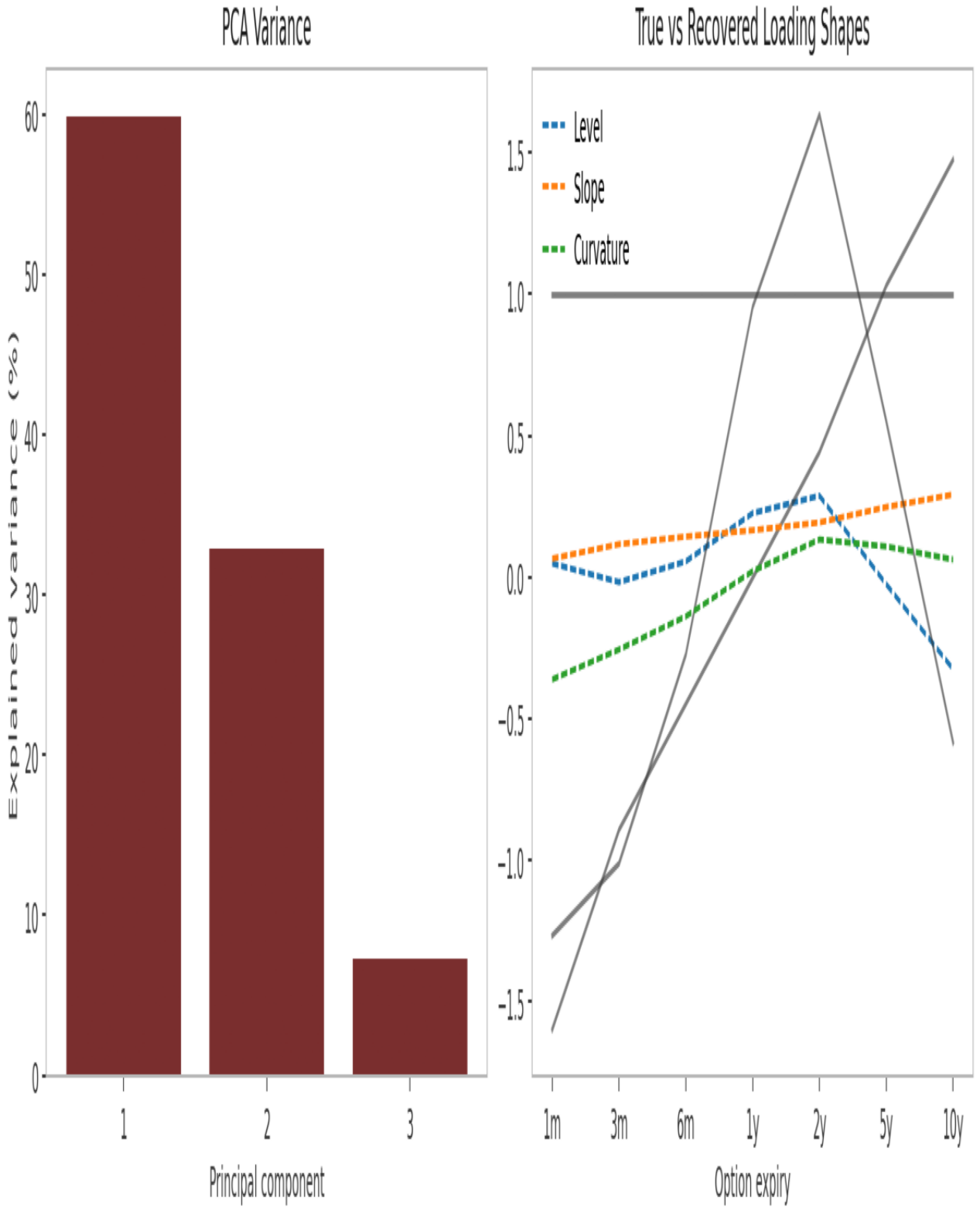
Three snapshots show the latent factor model moving the full cube, not isolated nodes.

Calibration Diagnostics



Residual heatmaps in vol bp for per-slice and cube-constrained fits.

PCA Recovery



Recovered loading shapes track the true level, slope, and curvature modes.

Arbitrage Violation Counts

	fit	violations	butterfly	calendar
Per-slice		21	16	5
Cube-constrained		22	16	6

Factor Recovery

PCA is run on simulated ATM log-vol changes. The first three components are aligned by correlation with the true loadings. This verifies that the analytics can see the factors the simulator injects.

Hedging Strategies

Delta-only removes underlying swap delta and lets vega run. Single-tenor vega-flat offsets

tenor-level raw vega into canonical liquid nodes. Bucketed vega-flat controls expiry

buckets. Factor-neutral solves a small quadratic hedge in level, slope, curvature, and skew

exposure.

Market-Maker Loop

Client trades arrive as a Poisson process. Cube point, side, strike offset, and notional are sampled. The client crosses the quote by construction, which is a simplification and gives the market-maker a clean spread-capture signal.

Quoting Overlay

The reservation vol follows an Avellaneda-Stoikov-style inventory charge in factor space.

This is a heuristic adaptation, not a full HJB derivation for a swaption cube. The value is

practical: widen where the next trade loads the already painful factor.

Results

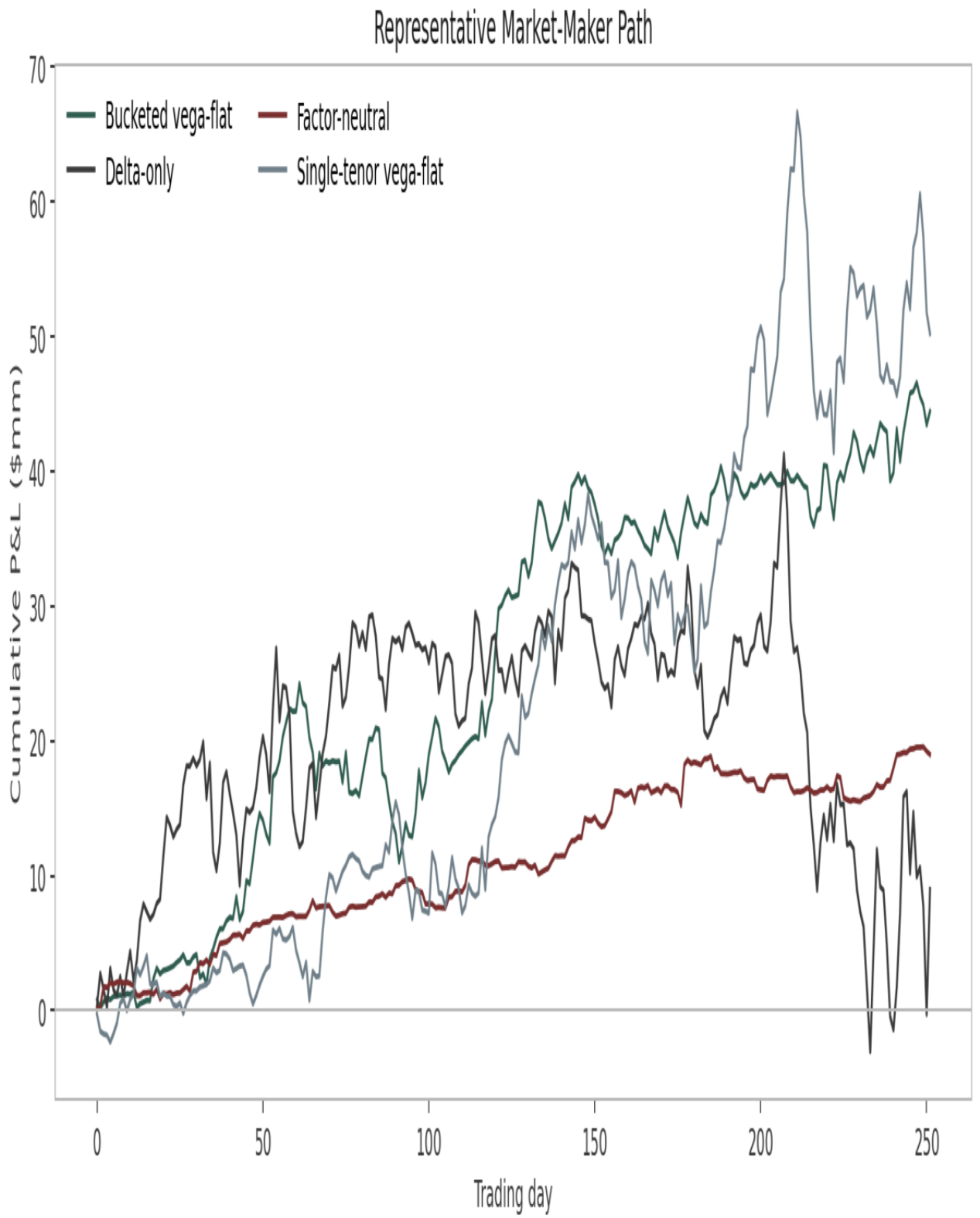
The factor-neutral hedge generally tightens the terminal distribution and reduces cross-vega

residuals. Bucketed hedging captures much of the improvement with less instrumentation.

The result is not free. Factor-neutral hedging trades more and therefore pays more hedge

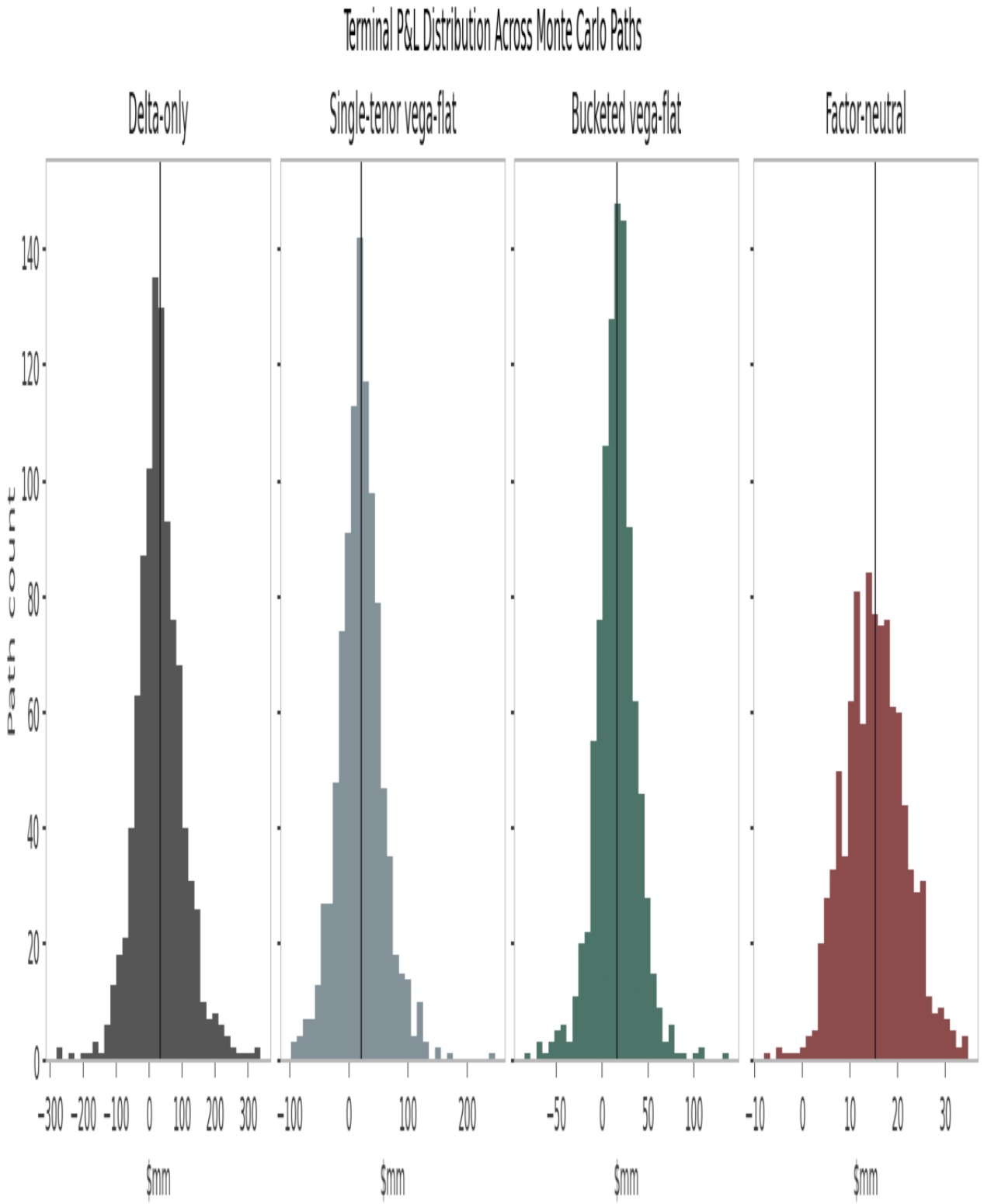
cost. That cost is visible in the attribution rather than smoothed away.

Representative Path



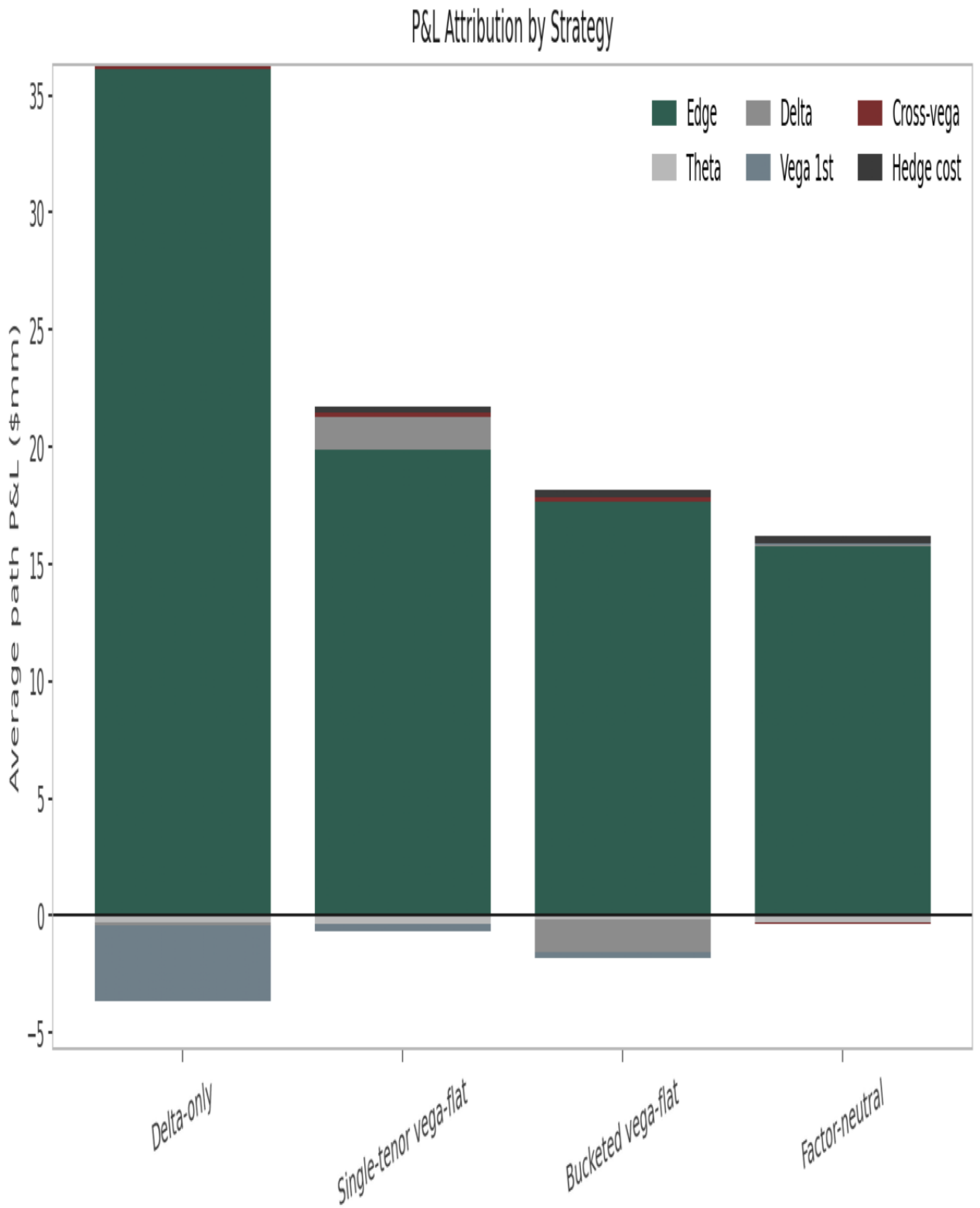
One Monte Carlo path illustrates how hedge rules diverge through time.

P&L Distributions



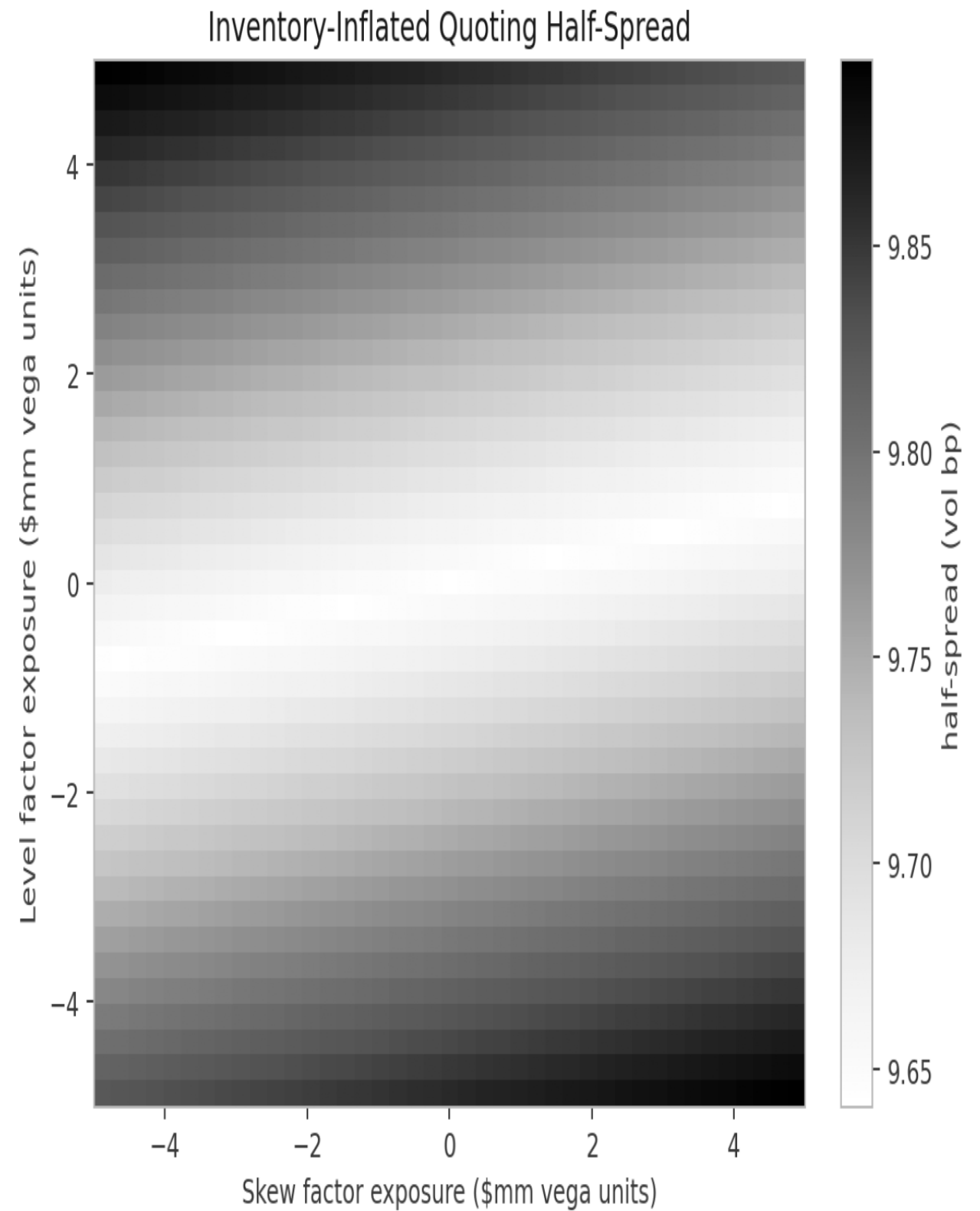
Terminal P&L distributions across all simulated market-making paths.

Attribution



Average path attribution exposes edge, theta, delta, vega, cross-vega, and hedge cost.

Quoting Heatmap



Half-spread rises with factor inventory in the quoted instrument direction.

Tail-Risk Table

	strategy	mean	std	var_5	cvar_5	sharpe
	Delta-only	32.64	70.38	-75.60	-115.65	0.46
	Single-tenor vega-flat	20.58	37.49	-41.45	-58.31	0.55
	Bucketed vega-flat	15.73	22.34	-19.81	-36.43	0.70
	Factor-neutral	15.27	6.42	5.06	2.71	2.38

Extensions

Natural extensions include a real-data overlay using exchange or broker marks, rough-vol factor dynamics, multi-curve discounting, cap-swaption joint calibration, Bermudan exposure via LSM, strategic IDB quoting, and an RL layer on top of the factor state.

References

Andersen and Piterbarg (2010); Hagan, Kumar, Lesniewski, and Woodward (2002); Rebonato

(2002); Bergomi (2016); Bartlett (2006); Avellaneda and Stoikov (2008); Gatheral (2006).